Information Literacy

Information Literacy

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Online Module Topic

- One of the Learning Goals in ENGR 133
- Definition:
  - The ability to know when information is needed and to be able to identify, locate, evaluate, and effectively use that information for the issue or problem at hand.

- Adapted from National Forum on Information Literacy

http://infolit.org/about-the-nfil/what-is-the-nfil/

Making Evidence-Based Decisions

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Online Module Topic

- Engineers make evidence-based decisions.
- To make evidence-based decisions, engineers must know how to ***identify, locate, evaluate, and effectively use information.***

This also applies to engineering students.

[Chemical engineer in the lab, online image]. Retrieved from http://chemicaleng.njit.edu/

Four Facets of Information Literacy

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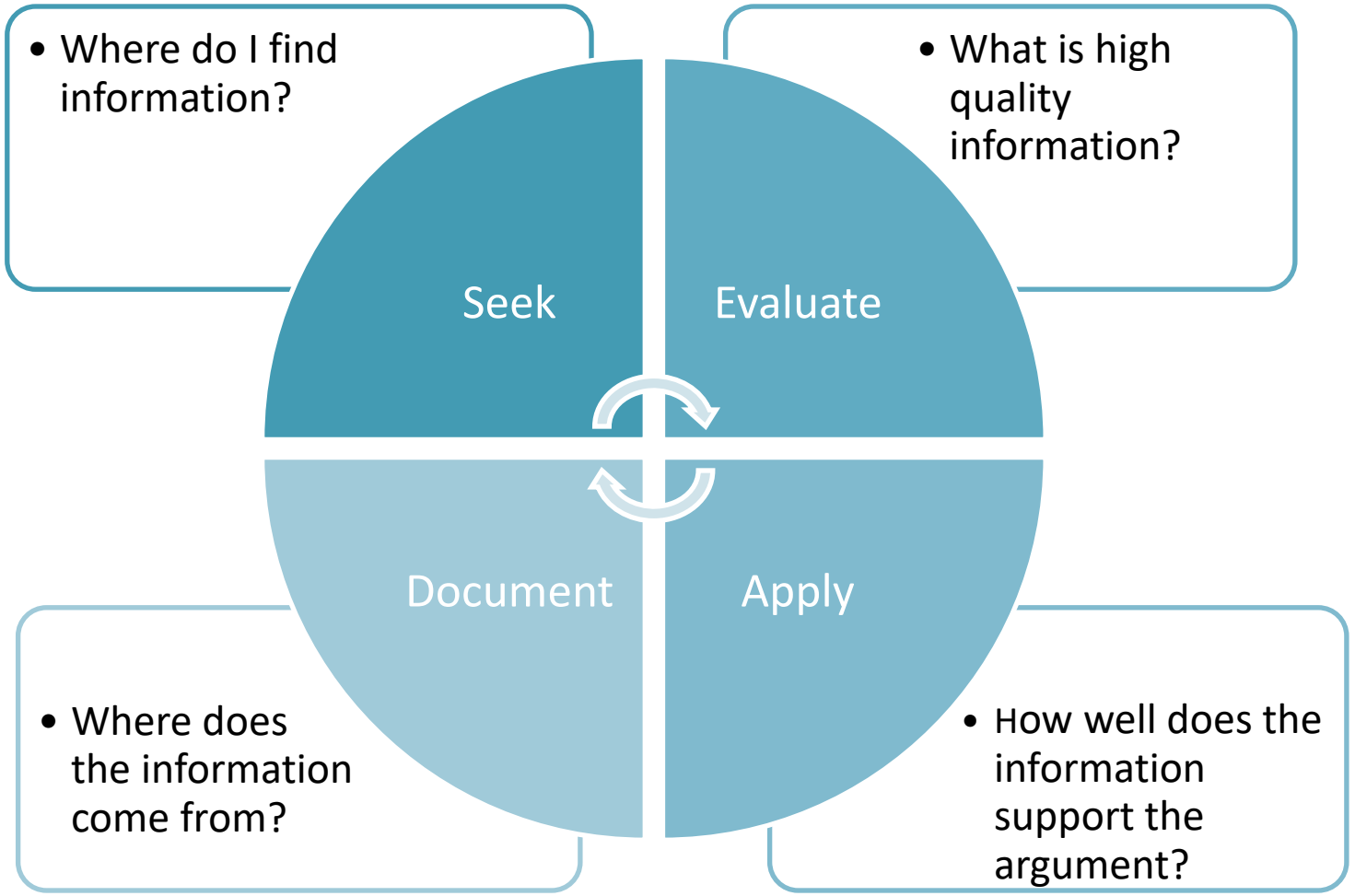
Online Module Topic



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Web Research: Purdue Libraries

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Seek

- Through the Purdue Libraries, you have access to
  - 3.5+ million print books
  - 1.2+ million electronic books
  - 4.7+ million articles from journals, reports, etc.

https://www.lib.purdue.edu/

or Google “Purdue libraries”

Web Research: Google Scholar

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Seek

- Search scholarly literature, including
  - Articles, theses, books, abstracts, etc. from academic publishers, professional societies, universities, and other websites

http://scholar.google.com/

Google Scholar: Your Library

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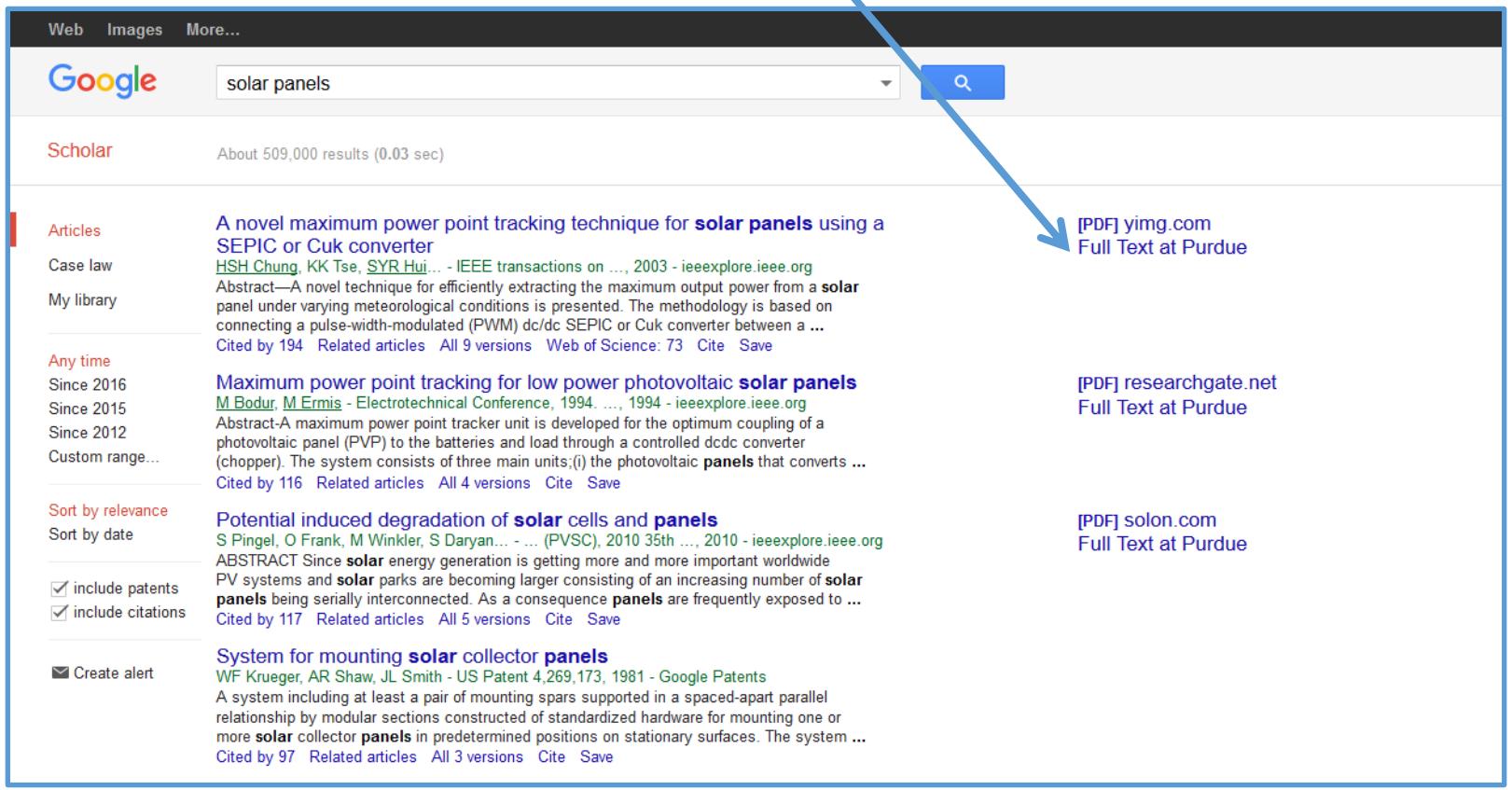
Seek

- Set your preferred library, such as Purdue:

  Settings > Library links > “Purdue”

Online Module Topic



Specialized Databases: COMPENDEX

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Seek

Engineering Village™
The first choice for serious engineering research.

Search | Selected records | Settings | Tags & Groups

New Search | View search history | Back to results | **1 of 1**

Full Text | Share | Print | Download

Abstract | Detailed

Highlight search terms

Record 1 from Compendex for: (((Keystone Pipeline) WN KY) AND ((impact on environment) WN KY)) , 1884-2017

Check record to add to Selected Records

Understanding state variability and recommendations for minimizing schedule risk in pipeline siting and development

Morton, John H.[1]; Rowe, Steve T.[2]; Hargrave, Rosemary C.[2]; Storey, Catherine F.[2]; White, Stephanie L.[2]

Source: *Proceedings of the Biennial International Pipeline Conference, IPC*, v 1, 2014; **ISBN-13:** 9780791846100; **DOI:** 10.1115/IPC201433504; **Conference:** 2014 10th International Pipeline Conference, IPC 2014, September 29, 2014 - October 3, 2014; **Sponsor:** Pipeline Division; **Publisher:** American Society of Mechanical Engineers (ASME)

Author affiliations:

[1] HDR, Omaha; NE, United States

[2] HDR, Denver; CO, United States

Abstract:

States are facing multiple challenges around new **pipeline** infrastructure as a result of unconventional oil and gas development. In many instances, federal oversight of interstate **pipelines** has failed to keep up with state interest in new **pipelines**. However, some states lack an efficient centralized or streamlined siting and permitting process to accommodate new development. This paper examines the different approaches by various states and what oil and gas companies can do to minimize risk by balancing the competing public and private interests to obtain the necessary permits and approvals without delaying project construction. The paper includes discussion of both Federal Energy Regulatory Commission (FERC) regulated natural gas **pipelines** as well as crude oil and product lines not regulated by FERC. In addition, this paper provides a firsthand perspective **on** how public opposition can change the regulatory landscape during the course of a project using the **Keystone** XL Project as an example. After the **Keystone** XL Project ran into permitting issues, the Nebraska Legislature modified its law, directing the state to evaluate the environmental, economic, social and other **impacts** of the Nebraska segment of the project. Because the state did not have a specific NEPA process, Nebraska had to first establish a process to conduct the state's evaluation and produce the functional equivalent of a NEPA document, resulting in project delays. This paper also describes how the evolving use of social media has provided a mechanism for rapid dissemination of information for both project opponents and proponents. There is a heightened national awareness that may now target certain projects based **on impacts** to select environmental resources such that just meeting the minimum regulatory requirements may not be enough moving forward. In today's regulatory **environment**, strategic planning and communications are critical to raise awareness in advance of the potential opposition.(6 refs)

Main heading: Laws and legislation

Controlled terms: Arches - Crude oil - Environmental regulations - Gas industry - Natural gas **pipelines** - Oil shale - Oil wells - Public utilities

Uncontrolled terms: Environmental resources - Federal energy regulatory commissions - Oil and gas companies - **Pipeline** infrastructure - Project construction - Regulatory **environment** - Regulatory requirements - Unconventional oil and gas

Classification Code: 408.2Structural Members and Shapes - 454.2Environmental Impact and Protection - 512.1Petroleum Deposits - 512.1.1Oil Fields - 522Gas Fuels - 523Liquid Fuels - 706Electric Transmission and Distribution - 971Social Sciences

Database: Compendex

Trustworthy Information Sources

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Evaluate

	Category	Resources
Internal	**Internal expertise**	❑ Discussions with clients and stakeholders ❑ Expert views and input ❑ Discussions with peers
	Research data	❑ Experimental data (experiments, models, simulations) ❑ User or process data (surveys, observations, images)
External	**Monographs**	❑ Handbooks, guides, standards, and manuals ❑ Technical reports ❑ Patents ❑ Statistical compilations ❑ Textbooks
	Periodicals	❑ Scholarly/journal articles ❑ News magazines, newspapers ❑ Trade magazines, popular magazines

Online Module Topic

Monograph: A scholarly piece of writing on a specific subject.
Periodical: A publication published at regular intervals..

(Purzer & Wertz, 2013)

Evaluating Web Information – Applying the CRAAP* Test

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Evaluate

- **Currency or Timeliness**
  - How important is it for your topic to have recent information?
  - When was the web page originally published, last updated, or revised?
  - Are all the links on the page functional? Can you determine if there is new information on the page?
- **Relevance**
  - How relevant to your topic is the information?
  - Can you find the same or better information from another source?
- **Authority**
  - What else has the author published about the topic?
  - Are there any credentials to establish them as expert?
  - Is it the main organization that provides information about a topic?
- **Accuracy**
  - Are there any sources cited for the information?
  - Can you verify any of the information in other independent sources or from your own knowledge?
- **Purpose**
  - Is the information fact or opinion?
  - Is it stating a point of view, promoting an idea, service or product?

*Modified version of the CRAAP Test created by Meriam Library at California State University-Chico

Is Wikipedia a Trustworthy Source?

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Online Module Topic

Evaluate

- Use Wikipedia as a starting point (if desired), but verify information by viewing cited sources directly or through additional research of your own.
- **Look up these secondary sources. Apply the CRAAP Test and cite the sources below if they pass the CRAAP test. Don't cite Wikipedia.**

References [edit]

1. ^ Coffee Plant *coffeeresearch.org Agriculture*
2. ^ What is the coffee tree? *Rombouts Coffee Shop-on-line*
3. ^ a b c d e f "10 Steps to Coffee". *National Association of Coffee U.S.A., Inc.* Retrieved 2007-12-24.
4. ^ http://www.casabrasilcoffees.com/learn/harvesting/methods-of-coffee-harvesting-selective-and-strip/
5. ^ a b c d e f g h "Coffee Reference Section". *Coffee Review*. Retrieved 2007-12-25.
6. ^ a b c "Field Processing". *International Coffee Organization*. Retrieved 2007-12-25.
7. ^ Nemerow, Nelson Leonard (1971). "Coffee Wastes". Liquid waste of industry: theories, practices, and treatment. Addison-Wesley Publishing Company.
8. ^ "Diverse coffees of Indonesia". *Specialty Coffee Association of Indonesia*. Archived from the original on 2008-08-02. Retrieved 2008-08-08.
9. ^ a b Marsh, T: "Review of the Aceh Coffee Industry", page 9. UNDP ERTR Livelihood Component, 2006
10. ^ "Coffee Mills: Density Sorting Coffee Beans". *Coffee Research Institute*. Retrieved 2007-12-25.
11. ^ http://www.sweetmarias.com/green.coffee.freshness.shtml
12. ^ http://www.sweetmarias.com/greenstorage.php
13. ^ http://www.coffeeshrub.com/shrub/content/about-vacuum-packaging
14. ^ http://www.ncausa.org/i4a/pages/index.cfm?pageid=74

Citations shown from the bottom of Wikipedia page on coffee production: https://en.wikipedia.org/wiki/Coffee_production

Evaluating All Information Sources

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Evaluate

Online Module Topic

- **Accuracy**
  - Can the information be verified in another source?
  - Are statements supported by evidence and/or citations?
  - Is the work edited by someone other than the author?
- **Authority**
  - Who is the author and what are his or her qualifications?
  - With what institutions or organizations is the author affiliated?
- **Currency**
  - When was the material published?
  - Is the date of publication appropriate?
- **Objectivity**
  - Is the author trying to sway your opinion or advocate a particular position?
  - Is the material presented based upon research or the author's opinion?

APPLY: How Well Does the Information Support the Argument?

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Online Module Topic

Apply

- Once information is found and evaluated, it needs to be applied to the given situation and used to support design decisions.
- The information must be **high quality.**
- The information must be **relevant.**
  - Sometimes information you gather may require you to change your decision or perspectives.

Four Facets of Information Literacy: Document

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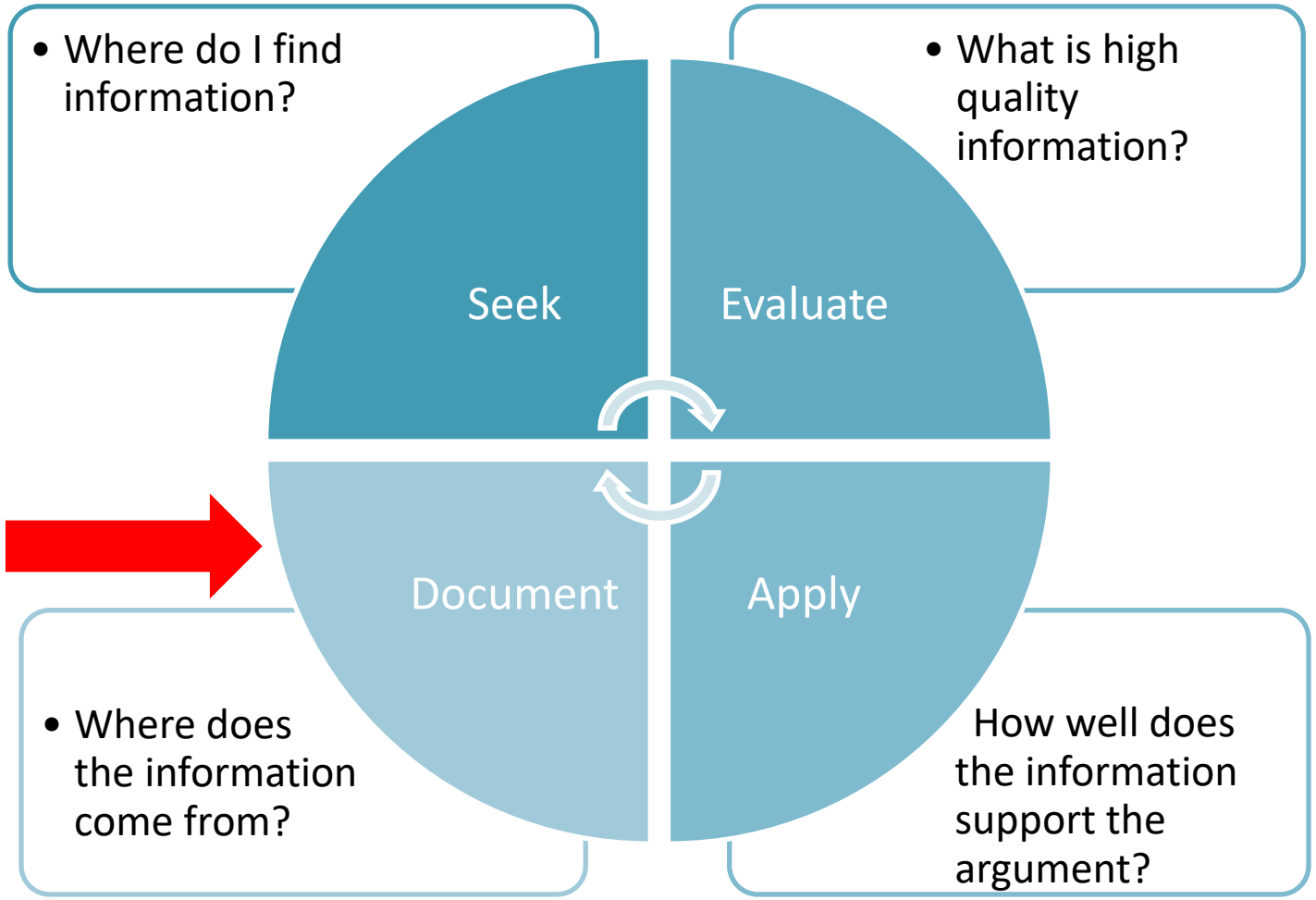
Document



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Citing Your Sources: When?

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- Both engineering students and practicing engineers must cite their sources of information.
- When are citations required?
  - **Required** to document words, ideas, or other productions that **originate somewhere outside of you**.
    - Information from books, journals, other printed sources
    - Web pages and other online sources
    - Pictures, artwork, diagrams, graphs
    - Quotes, interviews, emails
  - **Not required** for "common knowledge"

Online Module Topic

Document

But if in doubt, cite it!

Citing Your Sources: Why?

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Online Module Topic

Document

- Citations give credit to others for their words, ideas, research, etc.
- Citations allow readers to judge the quality of your information and the decisions made based on that information.
- Citations allow others to build on the information provided.

Not citing your sources can lead to allegations of plagiarism. Plagiarism is a form of academic dishonesty.

Plagiarism: The practice of taking someone else's work or ideas and passing them off as one's own.

Indiana University School of Education (2014). Definition of plagiarism. Retrieved from https://www.indiana.edu/~istd/definition.html
Purzer, S., & Wertz, R. (2014). Scaffold and access: Preparing students to be informed designers. In M Fosmire & D. Radcliffe (Eds.), *Integrating Information into the Engineering Design Process.* West Lafayette, IN: Purdue University Press.

Key Points on Citations

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Document

- Citations are required in ENGR 133 and will be required throughout your academic and professional life.
  - Know when citations are required.
  - Know the required style (APA, MLA, Chicago, etc.).
  - Use the available resources:

You are **not** required to memorize formats for citations. However, you should know **where to find the information you need.**

Information on Citations	Citation Generators
Purdue Online Writing Laboratory (OWL): http://owl.english.purdue.edu/owl/resource/560/01/ Bibme: http://www.bibme.org/citation-guide	Son of Citation Machine: http://citationmachine.net/index2.php Citefast: http://www.citefast.com/ Easybib: http://www.easybib.com/

Information Literacy – Key Points

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- **Seeking:** Look beyond Google for trustworthy information.
  - Use the Purdue Libraries and ask for help if you need it.
  - Use Wikipedia wisely.
- **Evaluating:** Evaluate information for accuracy, authority, currency, and objectivity.
- **Applying:** Accept that sometimes information you find may not be relevant or consistent.
  - Never misrepresent information to further your argument!
- **Documenting:** For each assignment, know the expectations for citations and references.
  - If in doubt, cite it!
  - Remember that failing properly cite your work is considered **plagiarism** and can lead to charges of **academic dishonesty**.

If these topics are new to you, consider reviewing the online module and other resources, since these concepts will apply to many of your classes at Purdue.